CNN-GP-Supplementary Material

I. CNN Module

The CNN in the devised CNN-GP is used as a feature-oriented representation learner for high-dimensional financial time-series data. Given that the original financial data is represented as X of $T \times N \times D$ where T, N, and D denote the number of trading days, the number of stocks, and the number of the raw features, respectively, CNN processes the multi-feature representation on each trading day and transforms the original D-dimensional feature space into a compact M-dimensional latent space Z of $T \times N \times M$.

As illustrated in Fig. S1, the CNN used in the devised CNN-GP consists of four consecutive 1×1 convolution blocks (Conv1D). The numbers of channels in the four Conv1D blocks are progressively expanded from 1 to 64, 64 to 128, 128 to 128, and 128 to 256, respectively. Besides, after the convolution layer in each of the four blocks, a batch normalization layer, a ReLU activation layer, a max pooling layer, and a dropout layer are progressively used for information extraction. After the four blocks, a global average pooling is adopted to obtain a compact 256-dimensional representation. Subsequently, an additional 1×1 Conv1D layer is used to map the 256-dimensional representation to M-dimensional latent features, which then serve as the input of the subsequent GP module for alpha factor mining.

To measure the quality of the M-dimensional latent features extracted by the CNN module, a classification layer is additionally employed to categorize the stocks into three coarse-grained groups, namely profitable, risky, and lossy, which are then compared with their true labels, to get the loss of CNN.

Here, it should be mentioned that the used CNN in this paper is very different from traditional temporal CNNs. Specifically, temporal CNNs slide convolution kernels across consecutive timestamps to capture short-term temporal dependencies for prediction. In contrast, the CNN used in this paper focuses on learning a compact representation of the high-dimensional stocks on each trading day, which is then used to mine the optimal alpha factor by GP.

II. Statistical Significance Analysis

To tell the performance difference between CNN-GP and the comparison methods, this paper conducts the two-sided Wilcoxon rank-sum test at the significance level 0.05 to compare the results of CNN-GP and that of each comparison method in terms of RankIC and ALER on the two datasets, namely CSI500 and CSI1000. Besides, the Friedman test at the same significance level 0.05 is also performed to obtain the overall performance ranking of each algorithm on the two datasets in terms of both RankIC and ALER.

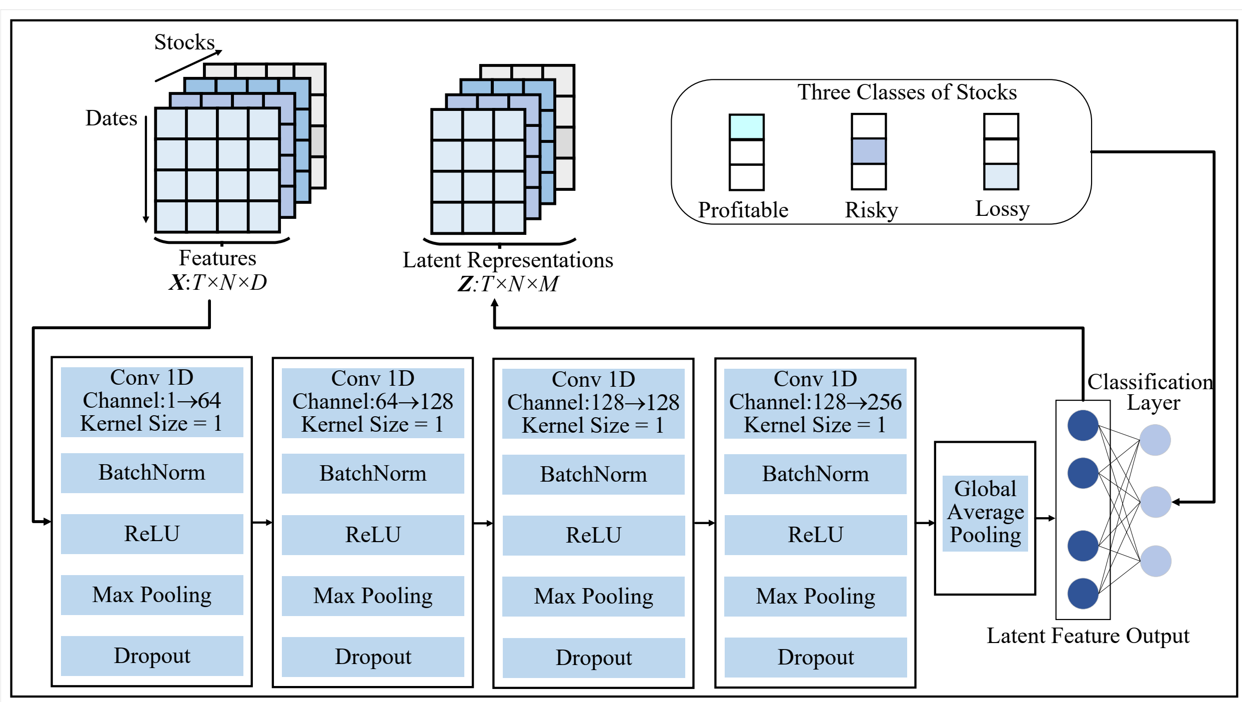

Fig. S1. The CNN architecture for latent feature extraction.

With the above two statistic tests, Table SI shows the comparison results between CNN-GP and the five comparison algorithms (AlphaForge, D-Va, QFR, RiskMiner, and SFAG), while Table SII displays the comparison results between CNN-GP and the four developed variants (CNN-XGBoost, CNN, GP, and XGBoost) on the two datasets. In these two tables, the symbols "+", "=", and "−" indicate that CNN-GP performs significantly better than, equivalently with, and significantly worse than the corresponding method, respectively.

As shown in Table SI, from the perspective of the p-values, in terms of both RankIC and ALER, CNN-GP significantly outperforms all the five comparison methods on both CSI500 and CSI1000. With respect to the Friedman rank, CNN-GP achieves significantly better overall performance than all the five comparison methods thanks to that its ranking is much lower than those of the five comparison methods.

As shown in Table SII, from the perspective of the p-values, concerning both RankIC and ALER, CNN-GP significantly outperforms all the four developed variants on both CSI500 and CSI1000. With respect to the Friedman rank, CNN-GP achieves the smallest ranking value among all methods. Besides, its ranking value is significantly smaller than those of the four developed variants. These demonstrates that CNN-GP obtains the best overall performance among all methods and its performance is significantly better than those of the four compared variants.

III. Runtime Analysis

This section investigates the computational cost of CNN-GP by comparing its runtime with those of the comparison methods. Specifically, we record the training time, the inference time, and the backtesting time for all methods. Then, the total time of each method by summing all the three kinds of time is also reported. Here, it should be mentioned that since we train all methods on all stocks and only use CSI500 and CSI1000 for test, the training time and the inference time, which occupy the most part of the execution time for each method, are the same for the two datasets. Therefore, we only report the computational time of all methods on one dataset, namely CSI500.

Figs. S2 shows the runtime of CNN-GP and the five comparison methods (namely AlphaForge, D-Va, QFR, RiskMiner, and SFAG) on CSI500, while Figs. S3 displays its runtime and those of the four developed variants (namely CNN-XGBoost, CNN, GP, and XGBoost) on CSI500.

As shown in Fig. S2, except for SFAG, the training time occupies the most part of the execution time for all the rest methods, while the inference time occupies the second most part of their execution time. Particularly, the training time of CNN-GP is

TABLE SI
Comparison results between CNN-GP and the five compared methods on the two datasets

Dataset / Metric	Quality	CNN-GP	AlphaForge	D-Va	QFR	RiskMiner	SFAG
CSI500 RankIC	Mean	**0.053020**	0.043300	0.034100	0.038000	0.021400	0.018800
	Std	**0.002953**	0.003305	0.003155	0.006678	0.004364	0.004195
	Median	**0.052650**	0.043600	0.034500	0.037800	0.021350	0.017050
	p-value	—	**2.46E-04+**	**1.83E-04+**	**2.46E-04+**	**1.83E-04+**	**1.83E-04+**
CSI500 ALER	Mean	**0.164800**	0.089200	0.062200	0.038010	0.071200	0.011000
	Std	**0.005587**	0.002858	0.003055	0.005625	0.003078	0.000606
	Median	**0.167200**	0.089250	0.062300	0.038500	0.071050	0.011100
	p-value	—	**1.82E-04+**	**1.82E-04+**	**1.82E-04+**	**1.82E-04+**	**1.80E-04+**
CSI1000 RankIC	Mean	**0.064600**	0.052300	0.043500	0.044300	0.033700	0.034600
	Std	**0.001678**	0.002877	0.003337	0.002677	0.003349	0.001478
	Median	**0.063950**	0.052350	0.043500	0.044200	0.033500	0.034950
	p-value	—	**1.83E-04+**	**1.83E-04+**	**1.83E-04+**	**1.83E-04+**	**1.83E-04+**
CSI1000 ALER	Mean	**0.157810**	0.063200	0.064620	0.061800	0.053000	0.082600
	Std	**0.002954**	0.002378	0.002726	0.003036	0.002997	0.002980
	Median	**0.157450**	0.063600	0.065050	0.061850	0.053050	0.083150
	p-value	—	**1.82E-04+**	**1.82E-04+**	**1.82E-04+**	**1.82E-04+**	**1.82E-04+**
Friedman Rank		**1.00**	2.50	3.75	4.00	5.00	4.75

Note: "Mean", "Std", and "Median" are the mean value, the standard deviation value and the median value over the 10 independent runs. The p-values are obtained from the two-sided Wilcoxon rank-sum test to compare CNN-GP with each comparison method at the significance level 0.05. "+", "=", and "−" indicate that CNN-GP performs significantly better than, equivalently with, and significantly worse than the corresponding method, respectively. The last row reports the average ranking obtained from the Friedman test.

TABLE SII

COMPARISON RESULTS BETWEEN CNN-GP AND THE FOUR DEVELOPED VARIANTS ON THE TWO DATASETS

Dataset / Metric	Quality	CNN-GP	CNN-XGBoost	CNN	GP	XGBoost
CSI500 RankIC	Mean	**0.053020**	0.045630	0.044400	0.042800	0.032810
	Std	**0.002953**	0.002443	0.006196	0.002680	0.003199
	Median	**0.052650**	0.045650	0.042200	0.043200	0.032850
	p-value	—	**4.40E-04+**	**9.11E-03+**	**1.83E-04+**	**1.83E-04+**
CSI500 ALER	Mean	**0.164800**	0.114100	0.105300	0.108100	0.019330
	Std	**0.005587**	0.002937	0.002985	0.002072	0.002536
	Median	**0.167200**	0.114150	0.105100	0.108050	0.019150
	p-value	—	**1.82E-04+**	**1.82E-04+**	**1.82E-04+**	**1.82E-04+**
CSI1000 RankIC	Mean	**0.064600**	0.059200	0.058400	0.051200	0.049100
	Std	**0.001678**	0.002851	0.001534	0.003222	0.002928
	Median	**0.063950**	0.059200	0.058700	0.051750	0.049750
	p-value	—	**1.83E-04+**	**1.83E-04+**	**1.83E-04+**	**1.83E-04+**
CSI1000 ALER	Mean	**0.157810**	0.098480	0.129000	0.077200	0.054040
	Std	**0.002954**	0.002795	0.000459	0.002848	0.002422
	Median	**0.157450**	0.098300	0.129000	0.077350	0.053450
	p-value	—	**1.82E-04+**	**1.80E-04+**	**1.82E-04+**	**1.82E-04+**
Friedman Rank		**1.00**	2.25	3.00	3.75	5.00

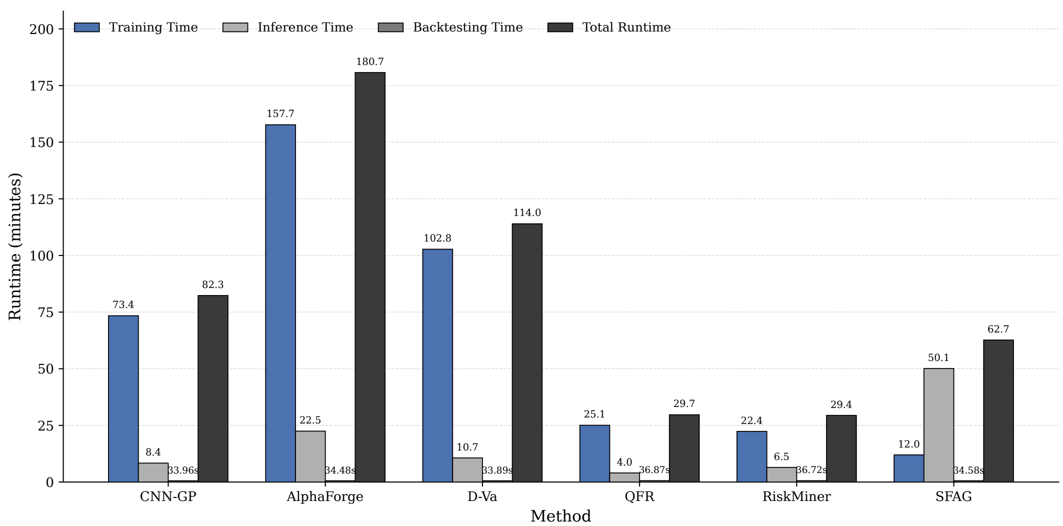

Fig. S2. Execution time including the training time, the inference time and the backtesting time of CNN-GP and the five comparison methods on CSI500.

much smaller than AlphaForge and D-Va, but larger than QFR, RiskMiner, and SFAG. In terms of the inference time, CNN-GP takes much less time than AlphaForge, D-Va, and SFAG, but more time than QFR and RiskMiner. Concerning the backtesting time, there are not so big difference among the six methods, but CNN-GP takes slightly smaller time than the five comparison methods.

As a whole, even though CNN-GP takes more time than some comparison methods, its performance is significantly better than all the five comparison methods regarding RankIC and ALER.

As shown in Fig. S3, regarding the training time, CNN-GP takes much less time than CNN-XGBoost, CNN, and GP, but more time than XGBoost. With respect to the inference time, CNN-GP consumes much less time than GP and XGBoost, but

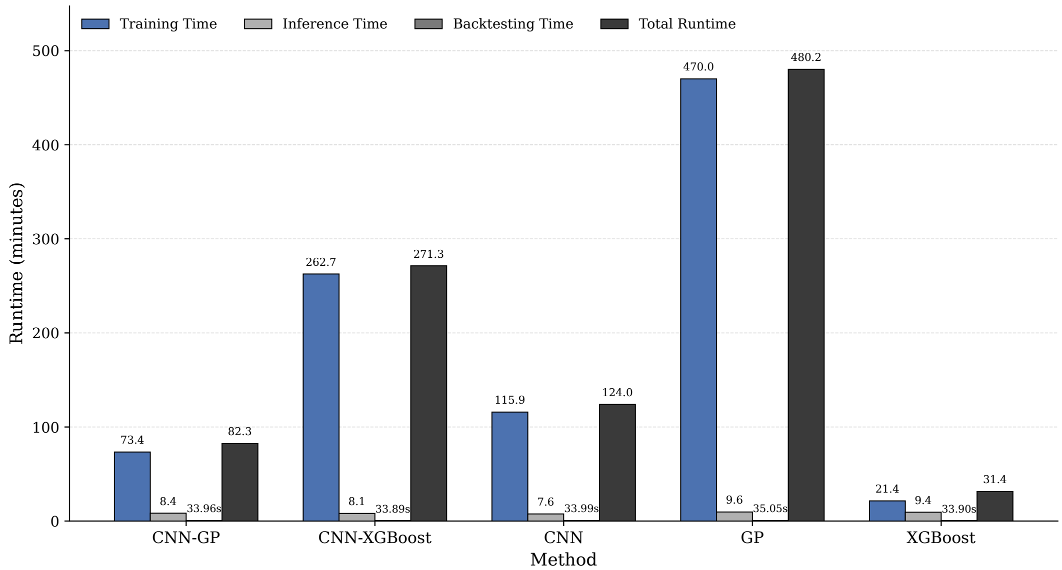

Fig. S3. Execution time including the training time, the inference time and the backtesting time of CNN-GP and the four compared variants on CSI500.

slightly more time than CNN-XGBoost and CNN. As for the inference time, there is no difference among all the five methods.

As a whole, CNN-GP takes much less time than CNN-XGBoost, CNN, and GP, but more time than XGBoost. However, its performance is significantly better than all the four comparison variants in view of both RankIC and ALER. These actually demonstrate the effectiveness of hybridizing CNN and GP for alpha factor mining and analyzing the financial time-series data.

IV. Sensitivity Analysis of CNN-GP to the Number of the Latent Features

This section investigates the influence of the number M of the latent features extracted by CNN on the performance of CNN-GP because such a number has significant influence on the alpha factor mining by the GP module. For this purpose, we offer different settings for M ranging from 8 to 30 with step size 2.

Fig. S4 presents the change curves of CNN-GP with the increase of the number of the latent features M on CSI500 and CSI1000 regarding RankIC, while Fig. S5 shows the change curves of CNN-GP with the increase of the number of the latent features M on CSI500 and CSI1000 regarding ALER.

From Fig. S4, in terms of RankIC, CNN-GP is not so sensitive to M since the gap of RankIC obtained by CNN-GP with different settings of M is not so large.

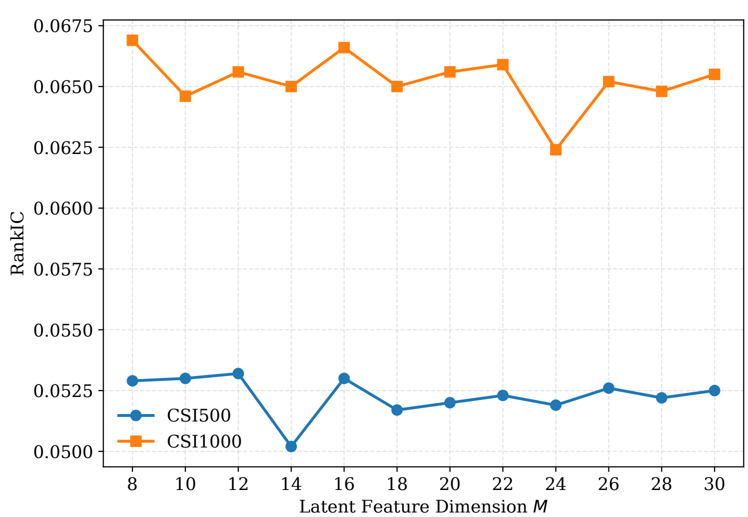

Fig. S4. Change curves of CNN-GP with the increase of the number of the latent features M on CSI500 and CSI1000 regarding RankIC.

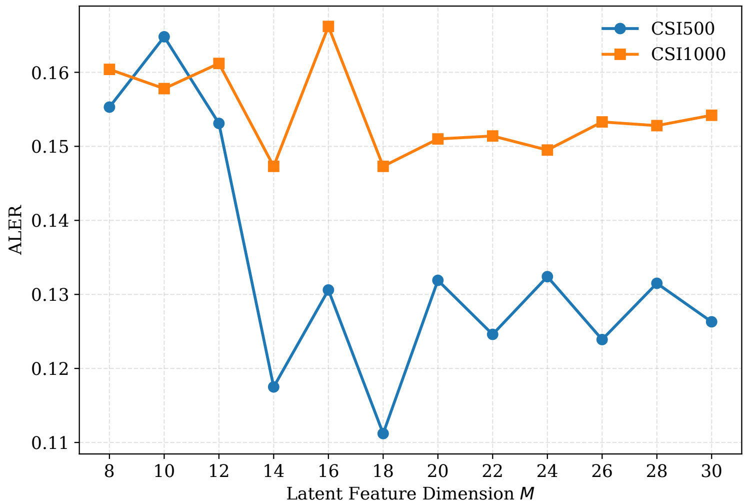

Fig. S5. Change curves of CNN-GP with the increase of the number of the latent features M on CSI500 and CSI1000 regarding ALER.

From Fig. S5, with respect to ALER, we find that the performance of CNN-GP differs obviously with different settings of M on the two datasets.

As a whole, it is observed that $M = 10$ helps CNN-GP gain the best overall performance on the two datasets. Therefore, $M = 10$ is set as the default setting for CNN-GP in this paper.